Module 4. Introduction to tagging and classification of events

Unit 4.1 What is it? Concept

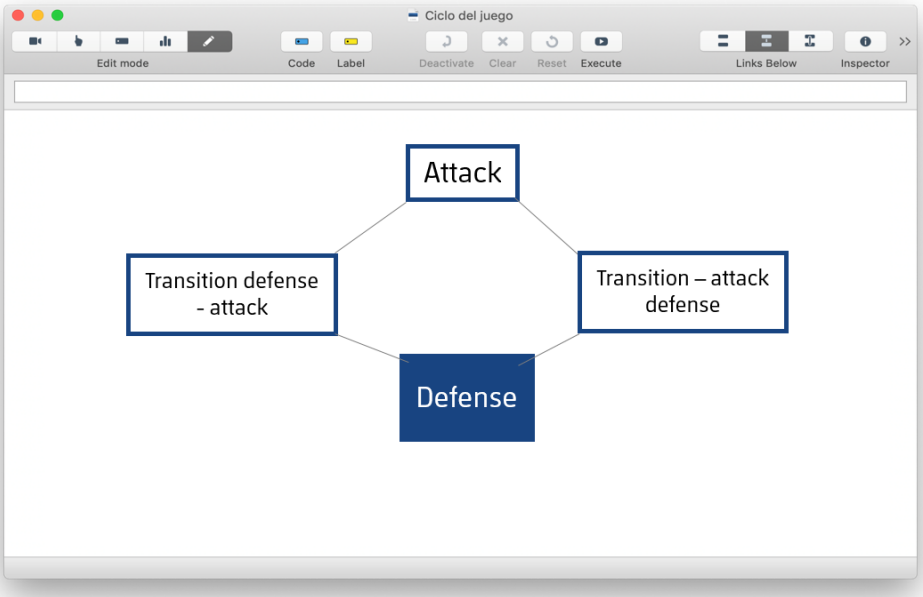
If we refer to a literal translation of tagging, we can find words such as monitoring, tracking or tags (WordReference.com, undated). This may involve a number of definitions that share a common link, if we look at their different meanings:

- Observe.
- Monitor.
- Classify.

With this in mind, we can define tagging as a process in which we classify a sequence of events by means of direct observation. In our field, this evidently refers to a sports event, and can be performed chronologically over time.

From this perspective, this is evidently carried out manually and for the sake of one or more people, based on previously established criteria. For example, we could discuss a general classification of the playing cycle itself.

Figure 1: Example of code window. Playing cycle. Generated with Hudl Sportscode



Source: Screenshot of Hudl Sportscode (Agile Sports Technologies, Inc., 2019).

This could serve as an example of general tagging that only includes information on certain points in the game. Through this window, and with the use of a software program, we could *tag* the different moments.

In order to have more descriptive and specific information about the concepts that we classify and how to generate them, we will explain below.

4.1.1 *Event data*

We cannot understand tagging without eventing, which can be generated by the analyst or taken from external sources. This data reveals actions performed in detail during play, focused on the ball. We can assign three characteristics to each event (Wetzler, 2017).

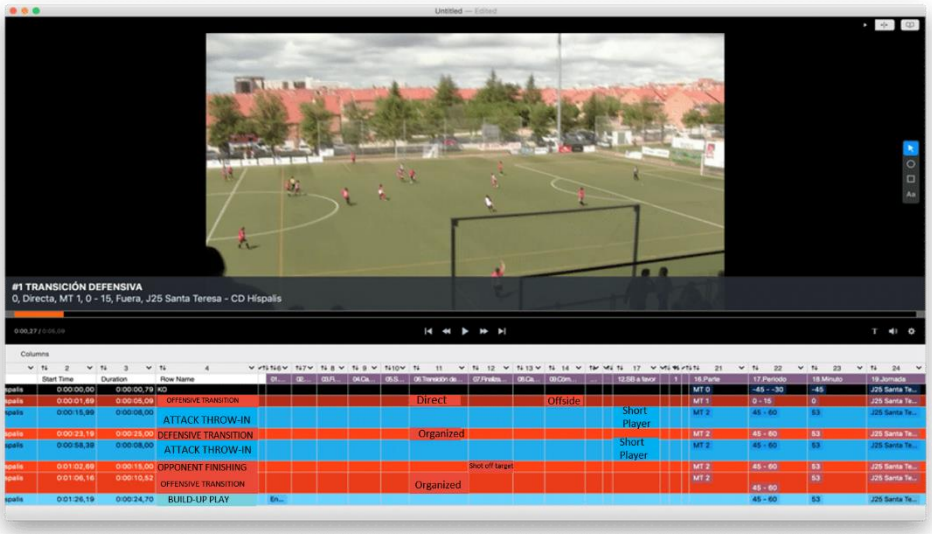
1. Action.
2. Timestamp.
3. Status.

The action refers to what happens (such as a kick), the timestamp - at what point in the game - (i.e. minute 15), along with the status of all information related to that action (player, context, competition, etc.).

One of the pioneering companies in eventing was Opta, which has been recording events since 2006, such as passes, shots or dribbling and their relation to time (Burn-Murdoch, 2018). This has expanded to the point of not only incorporating data, but also a prediction based on what could happen after the previous collection of various data in different contexts.

All of these resources have been a prelude to the development of patterns in the analysis of players, not only to assess individual or collective performance of one's own team or the opponents, but to also find out which players are preferable to others. One film that reflects this process is Moneyball (2011), in which a team is built based on objective statistical criteria, according to player performance. In order to see the results obtained by the team through this method of player acquisition, we recommend watching the movie.

Figure 2: Example of a sequence of events where each one reflects what happens on a tactical level, the circumstances of the action and the timestamp.



Generated with Sorter in Hudl Sportscode
Source: Screenshot of Hudl Sportscode (Agile Sports Technologies, Inc., 2019).

The combination of all of the elements that we can obtain through this event log can generate predictions, based on a number of algorithms that are constantly repeated. However, soccer is a constantly changing sport, in terms of teams and the shared space, so predictions are not always relevant.

Continuing along this line would mean discussing big data, which in turn would lead us away from the central theme of this module.

4.1.2 What type of data can we find?

There are many entities that either directly or indirectly provide services associated with the numerous events that occur in a soccer game, such as the media, betting agencies and clubs. Even now, when a game is broadcast, we can identify data related to the performance of a player or a team.

These resources can be found free of charge when it comes to professional soccer leagues. Therefore, they are an abundant and easily accessible resource for a detailed (inconclusive) analysis of what we can foresee in terms of a team's playing style or pattern.

Among the most typical data, which can easily provide us with enough information on the playing style and performance of a team, we have highlighted key examples:

- Goals. Made and received.

- Shots on goal. Made and received. Location and zone (outside the area).
- Shots off target Made and received. Location and zone (outside the area).
- Expected goals. Expected goals based on the quality of the kick, positioning, zone, type of kick, or type of pass that precedes it.
- Expected assists. Evaluates the possibility that a certain pass will become an assist.
- Sequences of passes in certain positions.
- Defensive coverage.

Figure 3: Sequence data of Costa's goal against Southampton during the 16/17 Premier



League season (Optasports, 2019).
Source: Screenshot of Optasports (Perform Group, 2019).

- Kickoffs / Interceptions / Fouls. Made and received.
- Dribbling.
- Possession and zones of most effective possession.
- Lanes where the game is distributed.
- Detailed player performance index with data such as goals, assists, key passes, minutes played, etc.
- Other elements.

All of these elements are associated with the point in time, the movement of the player or players, and even with the particular zone of the field. Evidently, this can provide us with

a wealth of information (even more so when the data is extracted from a greater number of games), but it is from external sources.

This information is generally provided by an external eventing, which is often supported by tracking data that is used in major soccer leagues. Its use is not decisive, but can be a useful addition for the analysis.

As a final reflection, all of these resources can provide a foundation for our analysis. However, personalized tagging based on the criteria of our philosophy, method or model, will determine (although not with the same objectivity) the direct relationship, based on what we analyze and what we want to achieve.

Unit 4.2 Tagging structure

In order to carry tagging in an efficient way, and at the same time organize the subsequent analysis, we need a number of resources, which we will explain in detail in the following sections.

Without these, everything we want to do will be difficult to store and to reuse in different situations.

4.2.1 Necessary resources

Audiovisual Resources

This may be evident, but it is necessary to clarify that we always need a video source for tagging, which will provide us with the visual resource to classify. The video of the competition can be selected from different sources, in accordance with the quality and angles that it provides.

We can find videos with television angles (broadcast), panoramic videos (also known as scouting feed), video from behind the goal or from a partial area of the field. Depending on what we want to analyze, we must choose the most suitable type of video.

Figure 4: Different signals taken from the same instant in a game. Generated with Hudl Sportscode



Source: Screenshot of Hudl Sportscode (Agile Sports Technologies, Inc., 2019).

In the event that we can only select one image, or if we are directly responsible for its recording, we will always obtain more information with a panoramic signal, in which we can observe the different behaviors and positions of the players, as well as their position in relation to the ball, the space and the opponent in the different lines.

Video Analysis Software

As we mentioned in the introduction, in order to obtain a better analysis and filtering of images based on the tagging selected, it will be necessary to use video analysis software. There are many available options with different features, prices and functions. Without wishing to determine which option might be best for users, we will conduct a brief review using the main options available.

Table 1: Summary of the main video analysis software programs on the market and their compatibility

Hudl Sportscode **Hudl**	hudlsportscode	*Mac*
Nacsport **Nacsport**	nacsport	*Window*
Longomatch **Fluendo**	LONGOMATCH BY FLUENDO	*Mac, Window, Linnux*
Eric **Eric Sports**	sports ERIC VIDEOANALYSIS SOFTWARE	*Window*
Vision **Catapult**	CATAPULT	*Mac*
Video Observer **Video Observer**	VIDEO OBSERVER REACH A HIGHER LEVEL	*Mac, Window*
Coach Capture **Chyronhego**	CHYRONHEGO	*Mac, Window*
Angles **Fulcrum**		*Mac*

Source: Prepared by the authors.

Skill and experience of the analyst

The vast majority of people who are not involved in analysis or who have not yet embarked on a professional path in this field may think that an analyst is a "video classifier or editor," which of course requires technical skill. If we take a video analysis software as our main point of reference, we will require a certain ability to use these tools, but they will only be a means to facilitate our actual analysis process.

With this in mind, we believe that an analyst should have the following key skills or characteristics:

- Knowledge of the game, its cycle and all of the related elements.
- Ability to interpret the game through video.
- Technical skill to be able to tag while interpreting what is happening, depending on how we observe the game.

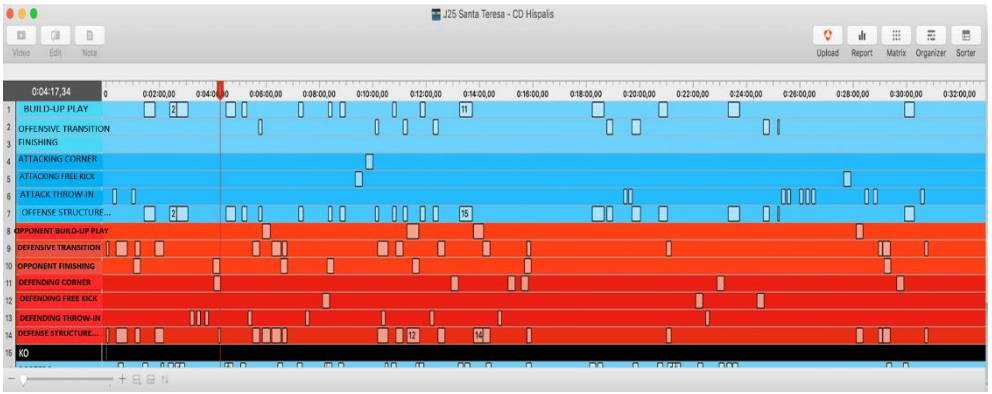
Evidently, all of these skills have one thing in common: practice. Clearly, this is something we can learn and develop. All those who choose this field can start out with this type of profession, taking into account that the level of efficiency will depend on prior experience and the amount of time devoted to this activity.

4.2.2 *Eventing*

The data presented by means of tagging can be represented in various ways and combinations. The most common classification method is based on codes and tags.

These **codes** and **tags** are assigned during or after a live game. They are indicated by certain points on a timeline.

Figure 5: Example of a timeline displaying all of the tags or clips for each concept. Generated with Hudl Sportscode



Source: Screenshot of Hudl Sportscode (Agile Sports Technologies, Inc., 2019).

In order to classify these, we must have a window where these codes and tags appear, which is known as the analysis panel, code window or push button panel. This will be the foundation of our analysis, through which we will classify all events.

All aspects of eventing that we observe in this timeline can be differentiated as follows:

Codes

This involves a succession or sequence at a given point in time in the analysis, and has a specific duration. The code will be given a specific name to describe the part of the game being analyzed. An example of a code could be **attack**, **possession** or **finishing**.

Codes can be applied automatically or manually with video analysis software. If it is manual, you only have to select this stage or moment in the game once we see and

interpret it, and pause again when it ends. This will create an event or video clip corresponding to that code.

Figure 6: Example of characteristics of a manual code (left) and an automatic code (right). Generated with Hudl Sportscode



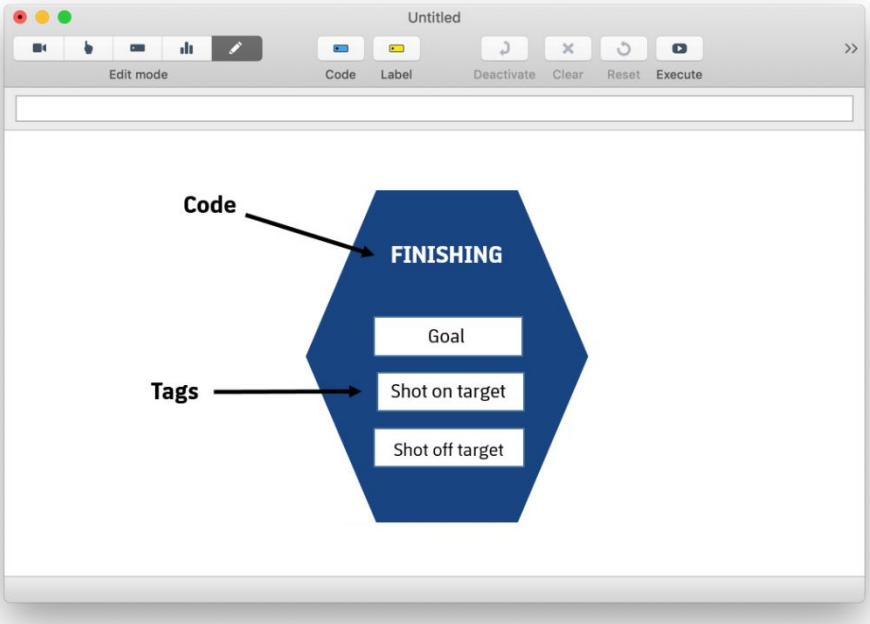
Source: Screenshot of Hudl Sportscode (Agile Sports Technologies, Inc., 2019).

On the contrary, for automatic code, once we interpret the action, we will only press it once. This generates an instant video clip. To do this, we will have to establish how much time we want for our video clip once pressed, with the option to define this before and after the clip is completed.

Tags

Creating a description through a tag is just as important as the categorization of a game concept through a code. With this tag, we will be able to characterize the code in a more detailed way. A simple example of this is as follows:

Figure 7: Example of code and tags on a push button panel. Generated with Hudl Sportscode



Source: Screenshot of Hudl Sportscode (Agile Sports Technologies, Inc., 2019).

In this regard, the same code, which represents an instant within a timeline (video footage of the game), could have different tags.

If we are able to categorize these codes with a number of tags, we will be able to, among other things:

- Quickly find the information we want.
- To be able to visualize it quickly.
- Take into account a number of items that help us to determine a pattern in our codes (events).
- Analyze using more solid concepts.
- Sort and reorder the information based on the criteria that we deem relevant.
- Customize our analysis as much as possible.

Therefore, prior to the analysis, it is essential to extensively categorize all of the elements that we believe represent the game.

Notes

The notes can be developed during or after the tagging process, which allows us to describe certain elements that we have not previously defined as codes or tags. An example of this might be:

Table 2: Notes during a tagging process

Code	Start of play
Tag	Short passes
Note	Formation of 3 at the start with wide centers, who move inside or center right.

Source: Prepared by the authors.

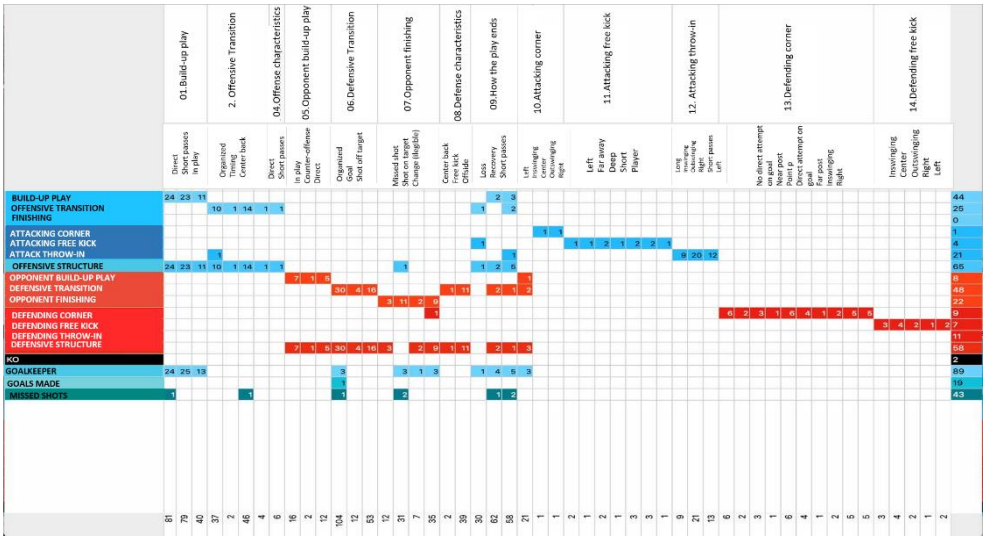
These notes would give us a more extensive and personalized analysis, which would allow us to develop new content based on the typical game behaviors of the team being analyzed.

4.2.3. Event matrix

With the codes and tags, we are able to classify all of the previously defined elements through tagging. We then have to be able to link them effectively, in order to work with all events. The matrix exists for this purpose. In it, we can quickly visualize how the data is cross-checked, based on the codes and tags.

Below is a visual example of where the codes appear to the left and the tags on top, representing the data cross-check in the central section; the eventing that is extracted based on our previous classification.

Figure 8: Example of an event matrix, extracted from the timeline in Figure 5. Generated with Hudl Sportscode



Source: Screenshot of Hudl Sportscode (Agile Sports Technologies, Inc., 2019).

This resource allows us to see the number of times that certain events occur, as well as to quickly visualize them by clicking on the corresponding data. At the same time, it is an element that we can use as a resource for statistical records, through analysis software and spreadsheet software (Excel, numbers).

On the other hand, by using the analysis software, we will be able to combine pre-established tags. For example, in this case, we could choose to combine two tags that can affect the **offensive transition** code, such as **direct** and **shot on goal**. This way, we will find all of the events of the aforementioned code in which the tags coincide.

Unit 4.3 Types of events based on certain moments in the game

The typology of events will provide us with a suitable analysis, based on our perspective and/or philosophy of the game. This classification is relevant, and will be shaped by aspects related to our playing model and that of the team we are analyzing (in the case of an opponent), or the type of playing style we analyze (competition or training).

However, we want to present a general classification that is as standardized as possible; based on the game cycle and which serves as a starting point for students. Evidently, this will be a simple point of reference from which we can build on, decrease or modify the categorization. We will divide the events based on the following key segments:

- Offense.
- Defense.
- Set pieces.

All of these events will be reflected in a window, where we can select each of these while watching the game. As was mentioned in the previous unit, this can be referred to as the analysis panel, code window or push button panels. It is similar to a blank sheet where we can create our vision of the game, and which will help us to classify and analyze the phenomena.

4.3.1. Offense

With this concept, we are referring to all of the situations in which the team being analyzed is in possession of the ball. This concept can be classified in multiple ways, based on:

- Speed of play.
- Progression in the game.
- Zone of the field.
- Patterns that are developed based on the above characteristics.
- Other aspects.

All of this can be summarized in specific events (codes), such as: start of play, organized attack, fast attack, direct attack, counterattack, or even more specific aspects such as the range and depth of play, changes in orientation or lateral centers.

The concepts are mere examples that can help guide our analysis. However, along with these concepts, the way that the game is understood largely depends on the criteria of the observer or analyst. Therefore, we prefer to use a more general classification.

4.3.2. Defense

Contrary to the offensive stage, this refers to all of the situations in which the team being analyzed is not in possession of the ball. While it is true that we can analyze situations in the possession stage, where the team uses offensive tactics for a better defense, but that is far too complex for the objective of this module.

The concept can be classified on the basis of:

- Height of the defensive block (low, medium or high block).
- Intensity and/or movements after losing possession of the ball.
- Intensity and/or movements in the different blocks.
- Zonal or individual defense.
- Other aspects.

These criteria could be described by events such as pressure, the zone and stage of returning to the defensive shape, defensive transition, or even more specific, such as pressure after loss of possession, marking in certain zones, or the location of the first line of pressure, among others.

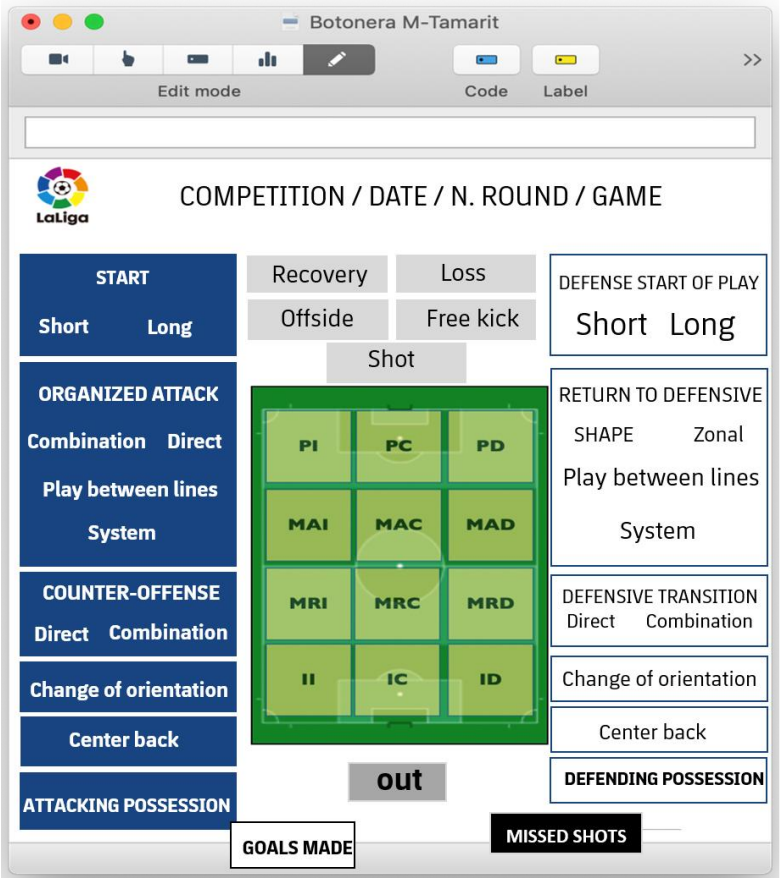
Bearing in mind all of these elements at the offensive and defensive level, we would like to give two examples of push button panels that could be useful for the analysis in general.

Figure 9: Example of a basic push button panel, with the main actions to be analyzed. Generated with Hudl Sportscode



Source: Screenshot of Hudl Sportscode (Agile Sports Technologies, Inc., 2019).

Figure 10: Example of an advanced push button panel, with the main actions to be analyzed. Generated with Hudl Sportscode



Source: Screenshot of Hudl Sportscode (Agile Sports Technologies, Inc., 2019).

4.3.3. Set pieces

Set pieces (SP) refer to all situations involving throw-ins, free kicks and corner kicks. In certain classifications, we could include kick-off. Unlike the offensive and defensive actions, set pieces are easier to classify because of their typology, and because they represent more "closed" playing situations. There is less room for interpretation with these.

Within the analysis, in order to categorize data into events and tags, we must be clear about what we want to know and how we will use that information. In doing so, we will take the following aspects into account.

Table 3: Aspects to be taken into account when categorizing set pieces

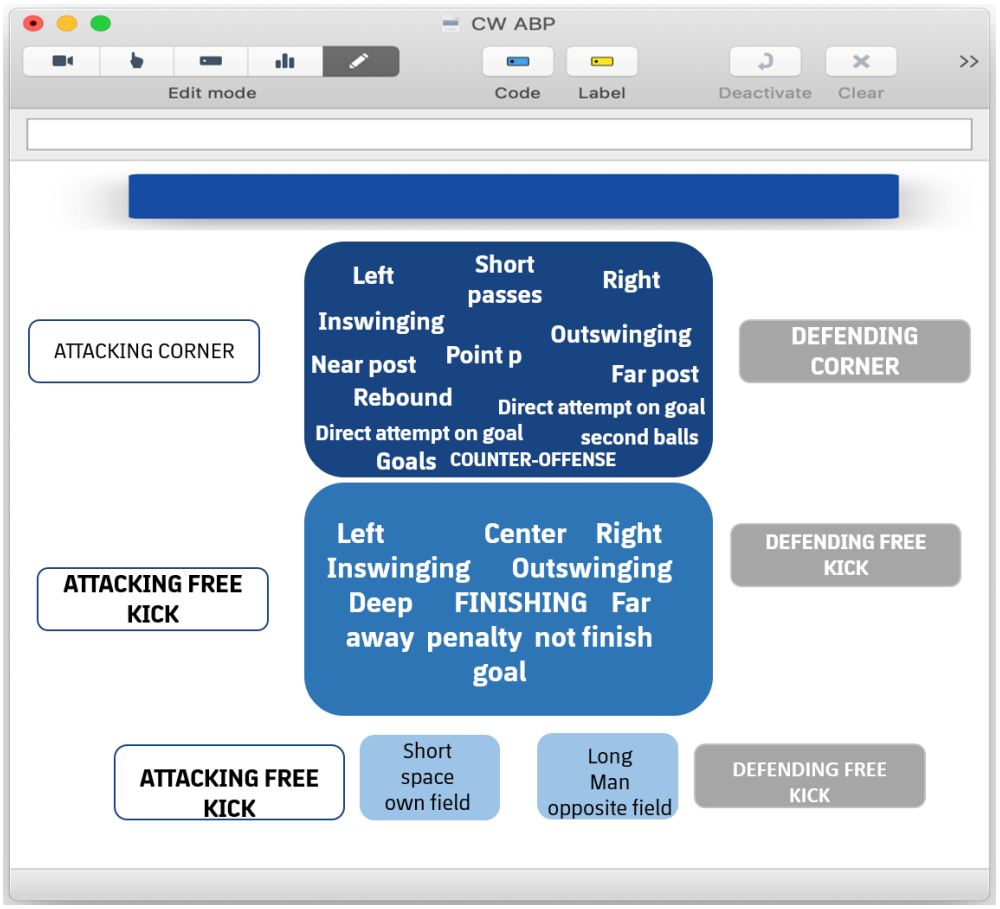
Set pieces in offense	Set pieces in defense
• Type of attack in set pieces • Who executes the direct attempt on goal (height, experience, aerial play, etc.). • Type of kick/throw-in. • Spaces where they are located. • Spaces to make a direct attempt on goal. • Players' movements. • Restarted plays. • Player kicking/throwing the ball in. • The area from which the free kick or throw-in is made.	• Type of defense. • Who is defending (height, experience, aerial play, etc.). • Type of marking. • Spaces where the players defend. • Volume of players defending. • The area from which the free-kick or throw-in is made.

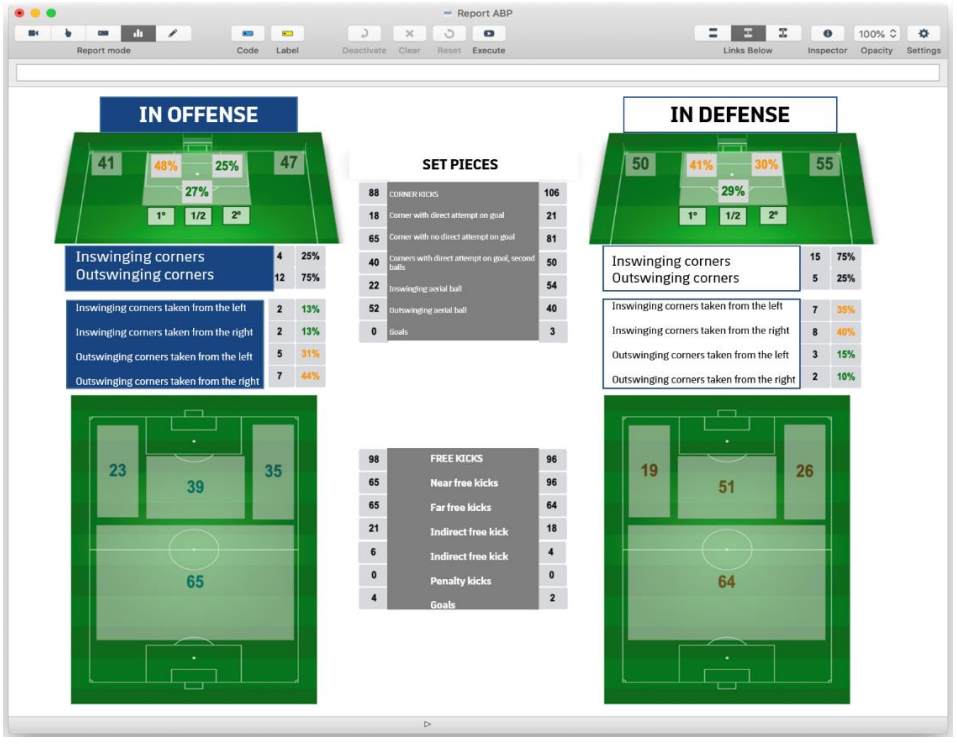
Source: Prepared by the authors.

These are some of the concepts that would help us in tagging these actions. Even so, there will always be numerous elements that are more open-ended, and can add a greater level of prediction to help us make or avoid a goal.

Taking into account some of the elements mentioned above, we will present an example of a push button panel to analyze set pieces, in addition to a report extracted from that analysis.

Figure 11: Push button panel and report on set piece in offense and defense. Generated with Hudl Sportscode





Source: Screenshot of Hudl Sportscode (Agile Sports Technologies, Inc., 2019).

4.3.4. Structure of tagging in training sessions

Training can be another element with which to perform tagging. Its classification will be somewhat different, since it is focused on the recording of tasks and their analyses, and based on other elements. Among these, we can find:

- Competitive period.
- Number of microcycle.
- Session number.
- Date of the session.
- Rival.
- Day of the week.
- Task type.
- Concepts developed.
- Other elements.

The combination of these, along with their development in codes and tags, will allow us to roughly classify (linked to video footage) everything we do during the year, at which point in the season, under what circumstances and against which rival.

Therefore, according to the daily tagging, this would help with:

- Session notes.
- Performance evaluation.
- Training database.
- Example in subsequent microcycles.
- Images that illustrate a playing model.
- Comparison with other periods of the year.
- Progress in the acquisition of technical-tactical and conditional skills.
- Other elements.

In this regard, we would complete the analysis cycle and implement a tagging of the entire video in the different points in time in the game, in order to truly ascertain if what we practice in training is what we bring to the competition.

References

Agile Sports Technologies, Inc. (2019). [Hudl Sportscode screenshot]. Lincoln.

Burn-Murdoch, J. (2018). *Cómo el análisis de datos ayuda a los clubes de fútbol a hacer mejores fichajes*. Retrieved from: https://www.ft.com/content/84aa8b5e-c1a9-11e8-84cd-9e601db069b8

De Luca, M., Horovitz, R. and Rudin, S. (2011). *Moneyball* [motion picture film]. United States. Scott Rudin Productions.

Perform Group. (2019). [Screenshot of Optasports]. Retrieved from: https://www.optasports.com/services/analytics/advanced-metrics/

Wetzler. M. (30 October 2017). *Event data: the root of all analyses*. https://tdwi.org/articles/2017/10/30/arch-all-event-data-root-of-all-analytics.aspx

Wordreference.com (s.f.). *Definición de Tagging*. Retrieved from: http://www.wordreference.com/es/translation.asp?tranword=tagging